

D861

# Electricity and Magnetism

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

In a demo using two small charged spheres on insulating stands bought from a science supplier, the force between them is 12 N when they are 3 cm apart. The spheres are painted silver. If the distance is increased to 6 cm, what is the force?

- A. 48 N
- B. 6 N
- C. 3 N
- D. 12 N, since the charges did not change.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2. [TRANSFER — reversed stem / different variable]

Two point charges exert a force  $F$  on each other. If both charges are doubled and the distance stays the same, what is the new force?

- A.  $8F$
- B.  $2F$
- C.  $4F$
- D.  $F/4$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

A lab on an upgraded electrostatics bench maps a uniform electric field pointing east between two plates. The plates are 20 cm tall and were cleaned before class. An electron is released between the plates. Which way does the force on it point?

- A. East.
- B. West.
- C. North, at right angles to the field.
- D. No force, since an electron's charge is too small to matter.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

A physics teacher who collects vintage electroscopes brings a negatively charged rod near, but not touching, a neutral metal sphere on an insulating stand. She touches the far side of the sphere with a grounded wire, removes the wire, then removes the rod. The room humidity is 35%. What is the final charge on the sphere?

- A. Negative.
- B. Positive.
- C. Neutral.
- D. Negative, because electrons jump from the rod through the air.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5.

A family's space heater, bought at a hardware store, is rated 1,500 W on 120 V and runs 4 hours a day for 30 days. Electricity costs \$0.15 per kWh. The heater is painted white. What does it cost to run for the month?

- A. \$2.16
- B. \$27
- C. \$0.90
- D. \$27,000

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6. [TRANSFER — reversed stem / different variable]

A 60 W bulb runs on 120 V. What current does it draw, and what is its resistance?

- A. 2 A and 60  $\Omega$
- B. 7,200 A
- C. 0.5 A and 60  $\Omega$
- D. 0.5 A and 240  $\Omega$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

At a junction on a classroom circuit board built by a robotics club, 5.0 A flows in through one wire. Two wires carry current out; one carries 2.0 A. The board has 12 LEDs mounted around its edge. What current flows in the second outgoing wire?

- A. 5.0 A
- B. 7.0 A
- C. 2.5 A
- D. 3.0 A

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

A student builds a parallel-plate capacitor from two aluminum sheets separated by plastic spacers; its capacitance is 40 pF. The sheets came from a recycled baking pan. If the plate separation is doubled while plate area stays the same, what is the new capacitance?

- A. 10 pF
- B. 40 pF, since the area did not change.
- C. 20 pF
- D. 80 pF

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9.

In a lab using a cathode-ray tube donated by a local TV repair shop, an electron beam travels straight through a region where a uniform magnetic field points in the same direction as the beam's motion. The tube is 30 cm long. What magnetic force acts on the electrons?

- A. Zero.
- B. A force opposite the motion that slows the beam.
- C. The maximum force, at right angles to the beam.
- D. A force along the field that speeds the electrons up.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**10.** [TRANSFER — reversed stem / different variable]

A proton ( $q = 1.6 \times 10^{-19}$  C) moves at  $2.0 \times 10^6$  m/s perpendicular to a 0.50 T magnetic field. What is the magnitude of the magnetic force?

- A.  $6.4 \times 10^{-13}$  N
- B.  $3.2 \times 10^{-13}$  N
- C. 0 N
- D.  $1.6 \times 10^{-13}$  N

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**11.**

A power company that serves 80,000 homes uses a step-down transformer with 2,000 turns on the primary and 100 turns on the secondary. The primary carries 2,400 V AC. The transformer is housed in a green metal box. What is the secondary voltage?

- A. 120 V
- B. 48,000 V
- C. 2,400 V, since transformers pass voltage through unchanged.
- D. 120 V, with the secondary current also 20 times smaller.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**12.** [TRANSFER — reversed stem / different variable]

A student connects a 12 V battery (steady DC) to a transformer's primary coil. After the first instant, what voltage appears across the secondary?

- A. Essentially zero.
- B. A voltage that keeps growing while the battery stays connected.
- C. 12 V, the same as the battery.
- D. A voltage set by the turns ratio, as with AC.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer C Concept 1

In a demo using two small charged spheres on insulating stands ~~bought from a science supplier~~, the force between them is 12 N when they are 3 cm apart. ~~The spheres are painted silver.~~ If the distance is increased to 6 cm, what is the force?

**Target word:** new force **Decides:** Distance doubles; force  $\propto 1/r^2$ .

|   |                    |                                                                  |
|---|--------------------|------------------------------------------------------------------|
| A | Inverted operation | Multiplies by 4; force falls as distance grows.                  |
| B | Adjacent category  | Uses a simple $1/r$ rule instead of $1/r^2$ .                    |
| C | <b>Correct</b>     | Doubling $r$ divides the force by $2^2 = 4$ : $12 \div 4 = 3$ N. |
| D | Dropped element    | Drops the effect of distance.                                    |

## 2. Answer C Concept 1 • transfer

Two point charges exert a force  $F$  on each other. If both charges are doubled and the distance stays the same, what is the new force?

**Target word:** force after changing charges **Decides:**  $F \propto q_1q_2$ .

|   |                    |                                                                       |
|---|--------------------|-----------------------------------------------------------------------|
| A | Confusion set      | Cubes the factor as if it were a volume scaling.                      |
| B | Dropped element    | Doubles only one of the charges.                                      |
| C | <b>Correct</b>     | $2 \times 2 = 4$ times the product of the charges.                    |
| D | Inverted operation | Applies the inverse-square rule to charge, which belongs to distance. |

## 3. Answer B Concept 2

A lab ~~on an upgraded electrostatics bench~~ maps a uniform electric field pointing east between two plates. ~~The plates are 20 cm tall and were cleaned before class.~~ An electron is released between the plates. Which way does the force on it point?

**Target word:** force direction on an electron **Decides:** Negative charge in an eastward field.

|   |                    |                                                               |
|---|--------------------|---------------------------------------------------------------|
| A | Inverted operation | East is the force on a positive charge.                       |
| B | <b>Correct</b>     | Force on a negative charge points opposite the field.         |
| C | Adjacent category  | That describes a magnetic-force pattern, not an electric one. |
| D | Absolute language  | Any charge in a field feels a force.                          |

## 4. Answer B Concept 3

A physics teacher ~~who collects vintage electroscopes~~ brings a negatively charged rod near, but not touching, a neutral metal sphere on an insulating stand. She touches the far side of the sphere with a grounded wire, removes the wire, then removes the rod. ~~The room humidity is 35%.~~ What is the final charge on the sphere?

**Target word:** sign of charge by induction **Decides:** Ground removed before the rod.

|   |                    |                                                                                                           |
|---|--------------------|-----------------------------------------------------------------------------------------------------------|
| A | Inverted operation | That is the result of touching the rod to the sphere (contact), not induction.                            |
| B | <b>Correct</b>     | The rod pushes electrons to ground; with the ground removed first, the sphere is left short of electrons. |
| C | Dropped element    | Drops the grounding step that let electrons leave.                                                        |
| D | Confusion set      | The rod never touches or sparks; charge moved to ground, not from the rod.                                |

## 5. Answer B Concept 4

A family's space heater, ~~bought at a hardware store~~, is rated 1,500 W on 120 V and runs 4 hours a day for 30 days. Electricity costs \$0.15 per kWh. ~~The heater is painted white.~~ What does it cost to run for the month?

**Target word:** monthly cost **Decides:**  $1.5 \text{ kW} \times 120 \text{ h}$ .

|          |                   |                                                                                                  |
|----------|-------------------|--------------------------------------------------------------------------------------------------|
| A        | Wrong stem number | Uses the 120 V rating as if it were 120 W of power.                                              |
| <b>B</b> | <b>Correct</b>    | $1.5 \text{ kW} \times (4 \times 30) \text{ h} = 180 \text{ kWh}$ ; $180 \times \$0.15 = \$27$ . |
| C        | Dropped element   | Drops the 30 days; that is one day's cost.                                                       |
| D        | Confusion set     | Uses watts as if they were kilowatts.                                                            |

## 6. Answer D Concept 4 • transfer

A 60 W bulb runs on 120 V. What current does it draw, and what is its resistance?

**Target word:** current and resistance from power **Decides:**  $I = P/V$ ;  $R = V/I$ .

|          |                    |                                                             |
|----------|--------------------|-------------------------------------------------------------|
| A        | Inverted operation | Divides V by P instead of P by V.                           |
| B        | Confusion set      | Multiplies power by voltage.                                |
| C        | Wrong stem number  | Uses the 60 W rating as the resistance.                     |
| <b>D</b> | <b>Correct</b>     | $I = 60/120 = 0.5 \text{ A}$ ; $R = 120/0.5 = 240 \Omega$ . |

## 7. Answer D Concept 5

At a junction on a classroom circuit board ~~built by a robotics club~~, 5.0 A flows in through one wire. Two wires carry current out; one carries 2.0 A. ~~The board has 12 LEDs mounted around its edge.~~ What current flows in the second outgoing wire?

**Target word:** missing branch current **Decides:** Current in = current out.

|          |                    |                                                                |
|----------|--------------------|----------------------------------------------------------------|
| A        | Confusion set      | Treats each branch as carrying the full current, as in series. |
| B        | Inverted operation | Adds instead of subtracting.                                   |
| C        | Adjacent category  | Assumes an even split, ignoring the given 2.0 A.               |
| <b>D</b> | <b>Correct</b>     | $5.0 - 2.0 = 3.0 \text{ A}$ .                                  |

## 8. Answer C Concept 6

A student builds a parallel-plate capacitor from two aluminum sheets separated by plastic spacers; its capacitance is 40 pF. ~~The sheets came from a recycled baking pan.~~ If the plate separation is doubled while plate area stays the same, what is the new capacitance?

**Target word:** capacitance after changing spacing **Decides:**  $C \propto 1/d$ .

|          |                    |                                                             |
|----------|--------------------|-------------------------------------------------------------|
| A        | Confusion set      | Applies an inverse-square rule borrowed from Coulomb's law. |
| B        | Dropped element    | Drops the dependence on separation.                         |
| <b>C</b> | <b>Correct</b>     | Doubling d halves C.                                        |
| D        | Inverted operation | Capacitance falls, not rises, with separation.              |

### 9. Answer A Concept 7

In a lab using a cathode-ray tube ~~donated by a local TV repair shop~~, an electron beam travels straight through a region where a uniform magnetic field points in the same direction as the beam's motion. ~~The tube is 30 cm long.~~ What magnetic force acts on the electrons?

**Target word:** force for parallel motion **Decides:**  $\theta = 0^\circ$ .

|   |                    |                                                                                        |
|---|--------------------|----------------------------------------------------------------------------------------|
| A | Correct            | $F = qvB \cdot \sin 0^\circ = 0$ .                                                     |
| B | Adjacent category  | Magnetic force is never along the velocity.                                            |
| C | Inverted operation | Maximum force needs perpendicular motion.                                              |
| D | Confusion set      | Magnetic forces never do work along the motion; this mixes in electric-field behavior. |

### 10. Answer D Concept 7 • transfer

A proton ( $q = 1.6 \times 10^{-19}$  C) moves at  $2.0 \times 10^6$  m/s perpendicular to a 0.50 T magnetic field. What is the magnitude of the magnetic force?

**Target word:** force magnitude **Decides:**  $F = qvB, \sin 90^\circ = 1$ .

|   |                    |                                                                                   |
|---|--------------------|-----------------------------------------------------------------------------------|
| A | Inverted operation | Divides by B instead of multiplying.                                              |
| B | Dropped element    | Drops the 0.50 T field.                                                           |
| C | Confusion set      | Zero applies to parallel motion, not perpendicular.                               |
| D | Correct            | $1.6 \times 10^{-19} \times 2.0 \times 10^6 \times 0.50 = 1.6 \times 10^{-13}$ N. |

### 11. Answer A Concept 8

A power company ~~that serves 80,000 homes~~ uses a step-down transformer with 2,000 turns on the primary and 100 turns on the secondary. The primary carries 2,400 V AC. ~~The transformer is housed in a green metal box.~~ What is the secondary voltage?

**Target word:** secondary voltage **Decides:**  $V_s = V_p \times N_s/N_p$ .

|   |                    |                                                             |
|---|--------------------|-------------------------------------------------------------|
| A | Correct            | $2,400 \times 100/2,000 = 120$ V.                           |
| B | Inverted operation | Uses the turns ratio upside down.                           |
| C | Dropped element    | Drops the turns ratio.                                      |
| D | Confusion set      | Current changes opposite to voltage; it is 20 times larger. |

### 12. Answer A Concept 8 • transfer

A student connects a 12 V battery (steady DC) to a transformer's primary coil. After the first instant, what voltage appears across the secondary?

**Target word:** transformer on DC **Decides:** No changing flux.

|   |                    |                                                                                  |
|---|--------------------|----------------------------------------------------------------------------------|
| A | Correct            | Steady current makes a steady field; with no change in flux, nothing is induced. |
| B | Inverted operation | Induced voltage depends on change, which stops once current is steady.           |
| C | Adjacent category  | The coils are not wired together; only changing flux links them.                 |
| D | Dropped element    | Drops the requirement that flux must change.                                     |

## Triage grid

Write each item number in the box it belongs to.

|                       | <b>Right</b>                                          | <b>Wrong</b>                                                                          |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| <b>Trap</b>        | <b>What it does</b>                                          | <b>In this drill</b> | <b>Your misses</b> |
|--------------------|--------------------------------------------------------------|----------------------|--------------------|
| Dropped element    | Leaves out a required term or step.                          | 8                    |                    |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 2                    |                    |
| Adjacent category  | The right idea in the wrong bucket.                          | 5                    |                    |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 11                   |                    |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 1                    |                    |
| Confusion set      | Every option is real; the test is exact role-matching.       | 9                    |                    |
| Camouflage         | Answers a stem fact that was never the question.             | 0                    |                    |

## Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                  | Rule                                                                                                                                                                                  | Trap                                                                                 | Tell                                                        |
|---|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1 | <b>Coulomb's law scaling</b>             | Electric force is proportional to the product of the charges and to $1/r^2$ : doubling distance quarters the force; doubling both charges quadruples it.                              | Treating force as dropping in step with distance.                                    | A force at one distance and a question about another.       |
| 2 | <b>Direction of electric force</b>       | The field points the way a positive charge would be pushed; a negative charge (electron) feels force opposite the field.                                                              | Assuming every charge moves along the field direction.                               | An electron placed in a field with a stated direction.      |
| 3 | <b>Charging by induction</b>             | Bringing a charged rod near a grounded conductor drives charge to or from ground; breaking the ground before removing the rod leaves the conductor with the opposite sign of the rod. | Expecting the same sign as the rod (that is charging by contact).                    | A rod held near but not touching, plus a ground connection. |
| 4 | <b>Electric power and energy cost</b>    | $P = VI = I^2R = V^2/R$ ; energy (kWh) = power (kW) $\times$ time (h); cost = kWh $\times$ rate.                                                                                      | Skipping the W-to-kW conversion or plugging voltage in as power.                     | A wattage, hours of use, and a price per kWh.               |
| 5 | <b>Kirchhoff's junction rule</b>         | Current into a junction equals current out; charge is conserved.                                                                                                                      | Assuming branches split current evenly or that each branch carries the full current. | A junction with some currents given.                        |
| 6 | <b>Parallel-plate capacitance</b>        | $C = \epsilon_0 A/d$ : capacitance rises with plate area and falls in proportion to plate separation.                                                                                 | Using an inverse-square rule or thinking separation does not matter.                 | A change in plate spacing or area.                          |
| 7 | <b>Magnetic force on a moving charge</b> | $F = qvB \cdot \sin\theta$ : maximum when motion is perpendicular to B and zero when motion is parallel to B.                                                                         | Expecting a force whenever a charge moves in a magnetic field.                       | The angle between velocity and field.                       |
| 8 | <b>Faraday's law and transformers</b>    | Only a changing magnetic flux induces a voltage; a transformer needs AC and sets $V_s/V_p = N_s/N_p$ , with current changing the opposite way.                                        | Expecting a transformer to work on steady DC, or current to scale with voltage.      | A turns ratio, a primary voltage, and the type of current.  |

**Course scope (public description):** Electricity and Magnetism is a comprehensive exploration of the fundamental principles of electricity and magnetism. This course is designed to help all students be successful in learning the core concepts of electricity and magnetism, and how they can be applied to real world technologies like electronics, motors, and infrastructure. Through a combination of theoretical instruction and experiments, students will build a robust foundation in electricity and magnetism, preparing them for a career as a physics teacher. This course emphasizes critical thinking, problem-solving, and the practical application of scientific principles, equipping students with the knowledge and skills necessary for success in the classroom.